

How Do Lindley Cases Arise?

A measure of evidence related to Bayes factors is the "evidence ratio" (Morey et al., 2016; Wagenmakers, et al. 2019), which can be written in two ways:

$$ER(\theta, x) = \frac{p(\theta|x)}{p(\theta)} = \frac{p(x|\theta)}{p(x)}. \quad (8)$$

The evidence-ratio compares the conditional probability of θ in the posterior distribution with its probability in the prior. If the same prior is used as the vague "alternative" against which to compute a Bayes factor for a point null, then the evidence-ratio at that point is the Bayes factor. Evidence-ratios greater than 1 favor the hypothetical θ , and taking this idea one step farther yields the concept of a "support interval" (Wagenmakers, et al. 2019), the values of θ for which $ER(\theta, x)$ exceeds some criterial value q . We may also take the analogous step of declaring a "rejection region" as the collection of values of θ for which $ER(\theta, x)$ falls below the criterial value $1/q$. Thus, support and rejection regions for θ are tantamount to regions displaying the point "nulls" with Bayes factors large or small enough to decide to retain or reject the "nulls".

There also is a straightforward connection between Bayes factors comparing a fixed point null with point-valued alternatives and the evidence-ratios for these alternatives. In equation (8), the denominator of the right-most ratio is a marginal likelihood and therefore a constant that does not involve θ . A BF that compares a point alternative θ against a point-null θ_0 may be written as

$$B(\theta, \theta_0, x) = \frac{p(x|\theta)}{p(x|\theta_0)}, \quad (9)$$

which therefore is proportional to the evidence-ratio for θ . Their ratio is

$$\frac{ER(\theta, x)}{B(\theta, \theta_0, x)} = \frac{p(x|\theta_0)}{p(x)} = ER(\theta_0, x), \quad (10)$$

the evidence-ratio for θ_0 . As mentioned above, if the prior involved in $ER(\theta, x)$ is the same as the vague alternative employed in a conventional point-null Bayes factor B_{01} , then $ER(\theta_0, x) = B_{01}$, regardless of the fact that $B(\theta_0, \theta_0, x) = 1$. Equation (10) implies that setting a criterial threshold for $ER(\theta_0, x)$ to determine a support region also sets a criterial threshold for $B(\theta, \theta_0, x)$, although the latter threshold is not known until the data have been collected. The converse also holds.

A source of utility in this connection is that both evidence-ratios and Bayes factors have clear and complementary betting interpretations. The evidence-ratio gives us the betting-odds $ER(\theta, x)$ on θ relative to the posterior distribution against the prior. The Bayes factor $B(\theta, \theta_0, x)$, on the other hand, gives the betting-odds on θ against θ_0 . Moreover, the relationship between $ER(\theta, x)$ and $B(\theta, \theta_0, x)$ provides an explanation for the Lindley-case syndrome.

Returning to the binomial example earlier in this paper, recall that H_0 is $\theta_0 = 1/2$ and data are $n = 985$ and $k = 524$ so the sample proportion is 0.532. Earlier we ascertained that $B_{01} = 3.344$. If we use the uniform prior then it also is the case that $ER(\theta_0, x) = 3.344$, and this is the evidence-ratio-Bayes-factor ratio in equation (10). If we set a threshold-value for evidence-ratios of $ER(\theta, x) > 3$ then the support region is $[0.499, 0.565]$. This interval includes $\theta_0 = 1/2$, and so we find that the data support the null. Now, recall that for hypothetical $0.51042 < \theta < 0.55349$, $B(\theta, \theta_0, x) > 3$ against the null. Thus, for θ in this interval the evidence ratios $R(\theta, x) > 3 * 3.344 = 10.032$. Therefore, the Lindley-case syndrome arises because the θ in $[0.51042, 0.55349]$ are much more strongly supported by the data than the null, $\theta_0 = 1/2$.

In our Gaussian example with $n = 5000$ and sample mean 0.035557 , we found a $B(0, \mu, \bar{X}, n) \geq 3$ against the null for comparisons with any specific μ in $[0.0068368, 0.0642769]$. In this example, $ER(\mu_0, x) = 3$ when $\mu_0 = 0$. The evidence-ratios for these interval limits therefore are 9. Again, the Lindley-case syndrome arises because the μ in $[0.0068368, 0.0642769]$ are more strongly supported by the data than $\mu_0 = 0$.

We now further examine the conditions under which Lindley cases occur. To begin, let us consider the range of μ values for which $B(0, \mu, \bar{X}, n) > q$ against the null at the same time as the Robert (2014) Bayes factor $R(\bar{X}, n) > q$ as a function of sample size. For any given $\bar{X}$ as n increases the $B(\bar{X}, 0, \mu, n) > q$ range also will expand and $R(\bar{X}, n)$ also will increase. We can see this by differentiating the second term on the right-hand side of equation (7), which gives

$$\frac{\partial \sqrt{n\bar{X}^2 - 2\log(q)}}{\sqrt{n}} \bigg/ \frac{\partial n}{n^{3/2} \sqrt{n\bar{X}^2 - 2\log(q)}} = \frac{\log(q)}{n^{3/2} \sqrt{n\bar{X}^2 - 2\log(q)}} \quad (11)$$

This is negative for $\log(q) < 0$ and positive for $\log(q) > 0$ when $n\bar{X}^2 > 2\log(q)$. Thus, when $B(0, \mu, \bar{X}, n) > q$ the limits grow further apart as n increases. Figure 2 illustrates this by comparing the $\log(B(0, \mu, \bar{X}, n))$ as a function of μ for $n = 5,000$ and $10,000$. The shallower curve is for the smaller sample size. As in Figure 1, the horizontal lines demarcate $\log(3)$ and $\log(1/3)$.